

SIMRET ARAYA GEBREEGZIABHER

Postdoc Fellow, Carnegie Mellon University, PA, USA

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RESEARCH INTERESTS

Human-AI Interaction; Cognitive Science and AI; Cognition-Inspired Interactive Systems; Human Learning, Reflection, and Decision-Making; Human-AI Alignment and Value Formation; Evaluation and Auditing of Agentic AI

PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow 06/2026–Present
Human-Computer Interaction Institute
Carnegie Mellon University

- Advisors: Prof. Haiyi Zhu, Dr. Yi-Chia Wang, Prof. Robert Kraut, and Prof. Sherry Wu
- Designing and evaluating AI-powered tools that support AI assisted and interactive human learning.

EDUCATION

Ph.D. in Computer Science and Engineering 01/2022–05/2026
University of Notre Dame Notre Dame, USA

- Advisor: Prof. Toby Jia-Jun Li
- Dissertation: *Cognition-Inspired Interactive Frameworks for Human-AI Alignment*
- Committee Members: Dr. Elena Glassman, Dr. Diego Gomez-Zara, Dr. Nitesh Chawla
- 🏆 IBM Ph.D. Fellow 2024–25

M.Sc. in Computer Science and Engineering 01/2022–08/2025
University of Notre Dame Notre Dame, USA

- Advisor: Prof. Toby Jia-Jun Li
- Major: Computer Science and Engineering

Bachelor of Science in Software Engineering 10/2016–10/2021
Addis Ababa Institute of Technology Addis Ababa, Ethiopia

- Major: Software Engineering
- Honor: Addis Ababa University 2021 Great Distinction

PAPERS UNDER REVIEW

[C.14] **Learning Through Interaction: Implicit versus Explicit Feedback in LLM-Simulated Customer Service Training**
Simret Araya Gebreegziabher, Jiayi Wang, Shiyu Zhang, Haiyi Zhu, Robert E Kraut, and Yi-Chia Wang
In Submission to CHI2027

[C.13] EvalAgent: Trajectory-Grounded Comparison of GUI Agent Behavior

Yukun Yang, **Simret Araya Gebreegziabher**, Hojun Yoo, Charles Chiang, Chaoran Chen, Annalisa Szymanski, Yinuo Yang, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gómez-Zará, and Toby Jia-Jun Li

In Submission to CHI2027

[C.12] CoBranchMR: Supporting Parallel Design and Conflict Resolution in Mixed Reality

Niloofer Sayadi, **Simret Araya Gebreegziabher**, Kaiyuan Tang, Yunhao Xing, Chaoli Wang, and Diego Gómez-Zará

In Submission to CHI2027

[C.11] Comparing Human Oversight Strategies for Computer-Use Agents

Chaoran Chen, Zhiping Zhang, Zeya Chen, Eryue Xu, Yinuo Yang, **Simret Araya Gebreegziabher**, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, and Toby Jia-Jun Li

In Submission to CHI2027

CONFERENCE PAPERS

[C.10] Stayin’Aligned Over Time: Towards Longitudinal Human-LLM Alignment via Contextual Reflection and Privacy-Preserving Behavioral Data

Simret Araya Gebreegziabher, Allison E Sproul, Yinuo Yang, Chaoran Chen, Diego Gomez-Zara, and Toby Jia-Jun Li

Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST 2026) [Paper]

[C.9] MultEval: Collaboratively Creating Criteria for LLM-as-a-Judge Systems

Charles Chiang, **Simret Araya Gebreegziabher**, Annalisa Szymanski, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Toby Jia-Jun Li, and Diego Gomez-Zara

Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work (CHI-WORK 2026) [Paper] [Code]

[C.8] The Behavioral Fabric of LLM-Powered GUI Agents: Human Values and Interaction Outcomes

Simret Araya Gebreegziabher, Yukun Yang, Charles Chiang, Dan Yoo, Chaoran Chen, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gomez-Zara, and Toby Jia-Jun Li

Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI 2026) [Paper] [Code]

[C.7] Dark Patterns Meet GUI Agents: LLM Agent Susceptibility to Manipulative Interfaces and the Role of Human Oversight

Jingyu Tang, Chaoran Chen, Jiawen Li, Zhiping Zhang, Bingcan Guo, Ibrahim Khalilov, **Simret Araya Gebreegziabher**, Bingsheng Yao, Dakuo Wang, Yanfang Ye, Tianshi Li, Ziang Xiao, Yaxing Yao, and Toby Jia-Jun Li

Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026) [Paper]

- [C.6] **Designing Staged Evaluation Workflows for LLMs: Integrating Domain Experts, Lay Users, and Model-Generated Evaluation Criteria**
 Annalisa Szymanski, **Simret Araya Gebreegziabher**, Oghenemaro Anuyah, Ronald A. Metoyer, and Toby Jia-Jun Li
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)
 [Paper]
- [C.5] **Supporting Co-Adaptive Machine Teaching through Human Concept Learning and Cognitive Theories**
Simret Araya Gebreegziabher, Yukun Yang, Elena L. Glassman, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2025)
 [Paper]
 **Best Paper Award**
- [C.4] **Leveraging Variation Theory in Counterfactual Data Augmentation for Optimized Active Learning**
Simret Araya Gebreegziabher, Kuangshi Ai, Zheng Zhang, Elena L. Glassman, and Toby Jia-Jun Li
Findings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Findings 2025) [Paper] [Code]
- [C.3] **MetricMate: An Interactive Tool for Generating Evaluation Criteria for LLM-as-a-Judge Workflow**
Simret Araya Gebreegziabher, Charles Chiang, Zichu Wang, Zahra Ashktorab, Michelle Brachman, Werner Geyer, Toby Jia-Jun Li, and Diego Gomez-Zara
Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work (CHI-WORK 2025) [Paper]
- [C.2] **The Obvious Invisible Threat: LLM-Powered GUI Agents' Vulnerability to Fine-Print Injections**
 Chaoran Chen, Zhiping Zhang, Bingcan Guo, Shang Ma, Ibrahim Khalilov, **Simret Araya Gebreegziabher**, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, and Toby Jia-Jun Li
(ACM Transactions on AI Security and Privacy (TAISAP) 2026) [Paper]
- [C.1] **PaTAT: Human-AI Collaborative Qualitative Coding with Explainable Interactive Rule Synthesis**
Simret Araya Gebreegziabher*, Zheng Zhang*, Xiaohang Tang, Yihao Meng, Elena L. Glassman, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2023)
 [Paper] [Code] [Pip]

WORKSHOP PAPERS, DEMOS, AND EXTENDED ABSTRACTS

- [L.8] **Steering Vision Models towards Subjective Concepts with Language**
 Mihret Bekele, Dawit Getahun Mengistu, Kidus Paulos, Natnael Abayneh, Amanuel Gizachew Abebe, and **Simret Araya Gebreegziabher**
CHI 2026 BiAlign Workshop

- [L.7] **EvalAgent: Interactive Comparative Evaluation of Computer-Using GUI Agents**
Yukun Yang, **Simret Araya Gebreegziabher**, Dan Yoo, Charles Chiang, Chaoran Chen, Annalisa Szymanski, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gomez-Zara, and Toby Jia-Jun Li
IUI 2026 Demo
- [L.6] **MultEval: Collaboratively Creating Criteria for LLM-as-a-Judge Systems**
Charles Chiang, **Simret Araya Gebreegziabher**, Annalisa Szymanski, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Toby Jia-Jun Li, and Diego Gomez-Zara
IUI 2026 Demo
- [L.5] **Toward a human-centered evaluation framework for trustworthy llm-powered gui agents**
Chaoran Chen, Zhiping Zhang, Ibrahim Khalilov, Bingcan Guo, **Simret A Gebreegziabher**, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, Toby Jia-Jun Li
CHI 2025: HEAL Workshop
- [L.4] **MetricMate: An Interactive Tool for Generating Evaluation Criteria for LLM-as-a-Judge Workflow**
Simret Araya Gebreegziabher, Charles Chiang, Zichu Wang, Zahra Ashktorab, Michelle Brachman, Werner Geyer, Toby Jia-Jun Li, and Diego Gomez-Zara
IUI 2025 Workshop on Human-AI Co-Creation with Generative Models (HAI-GEN)
- [L.3] **MOCHA: Model Optimization through Collaborative Human-AI Alignment**
Simret Araya Gebreegziabher, Elena L. Glassman, and Toby Jia-Jun Li
Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST 2024 Adjunct) [Paper] [Video]
- [L.2] **CoCo Matrix: Taxonomy of Cognitive Contributions in Co-writing with Intelligent Agents**
Ruyuan Wan, **Simret Araya Gebreegziabher**, Toby Jia-Jun Li, and Karla Badillo-Urquiola
Extended Abstracts of the 16th ACM Conference on Creativity and Cognition (C&C 2024)
- [L.1] **A Taxonomy for Human-LLM Interaction Modes: An Initial Exploration**
Jie Gao*, **Simret Araya Gebreegziabher***, Kenny Choo, Toby Jia-Jun Li, Simon Perrault, Thomas Malone
Extended Abstracts of the 2024 ACM Conference on Human Factors in Computing Systems (CHI EA 2024)

RELEVANT RESEARCH EXPERIENCE

Graduate Research Assistant

01/2022–05/2026

University of Notre Dame

Advisor: Toby Li

- Developed cognition-inspired human–AI systems supporting human learning, qualitative analysis, preference elicitation, and collaborative model evaluation
- Led mixed-method experiments and longitudinal deployments examining how people learn from, evaluate, and revise judgments about AI systems
- Combined cognitive theory, interactive system design, behavioral analysis, qualitative methods, and machine-learning evaluation across interdisciplinary research projects

AI/ML Research Intern

05/2025–08/2025

*Apple**Host: Tim Muss*

- Conducted a large-scale user research with Apple internal participants to understand the human input in LLM evaluation
- Devised a mixed methods approach by combining human factors and user research lens
- Presented results and actionable insights to senior directors and leadership

Research Scientist Intern

05/2023–11/2023

*Meta Reality Labs**Host: Yan Xu*

- Worked on Human-LLM interaction methods to support data exploration in UX-research, and qualitative research
- Developed and deployed an interactive system that is being used as an internal research tool
- Wrote a manuscript and submitted to an HCI conference

Research Assistant

05/2020–02/2021

*University of Michigan**Host: Valeria Bertacco*

- Worked as a research assistant at the University of Michigan with Professor Valeria Bertacco as my faculty advisor. I worked on computer architecture projects namely Composable Benchmarking and redesigning memory access for faster explainable-AI algorithms.

TEACHING EXPERIENCE**Teaching Assistant, CSE 40748/60748: Human-AI Collaborative Systems**

Spring 2025

*University of Notre Dame***Teaching Assistant, CSE 40424: Human Computer Interaction**

Fall 2022

*University of Notre Dame***Teaching Assistant, CSE 30321: Computer Architecture**

Spring 2022

*University of Notre Dame***Instructor, Girls Can Code**

Summer 2018

*X-hub Ethiopia***STUDENTS MENTORED****Allison E Sproul** (University of Notre Dame)

Spring 2026

- *Worked on longitudinal user intent detection.* [C.10]

Mihret Bekele (Addis Ababa University)

Fall 2025

- *Worked on detecting subjectivity in image description and steering image retrieval models.* [L.8]

Dawit Getahun Mengistu (Addis Ababa University)

Fall 2025

- *Worked on detecting subjectivity in image features and using sparse autoencoders for model steering.* [L.8]

Kidus Paulos (Addis Ababa University)	Fall 2025
• <i>Worked on detecting subjectivity in image description and steering image retrieval models.</i> [L.8]	
Yukun Yang (Tongji University)	Summer 2024, 2025
• <i>Worked on LLM and LLM-based GUI agent evaluation.</i> [L.7][C.5][C.8]	
Dan Yoo (University of Notre Dame)	Spring 2025
• <i>Worked on generating value-based GUI agent web browsing traces.</i> [C.8]	
Zichu Wang (Carnegie Mellon University)	Summer 2024
• <i>Worked on generating user friendly interfaces to represent LLM-as-a-judge outcomes</i> [L.4][C.3]	
Ava DeCroix (University of Notre Dame)	Summer 2022
• <i>Worked on data collection on volunteer-based disaster relief organizations.</i>	

GRANTS & AWARDS

Best Paper Award: CHI 2025 [C.5]	2025
IBM Ph.D. Fellowship (40,000 USD)	2024–2025
Notre Dame–IBM Tech Ethics Hackathon: Second Place	2024
Gary Marsden Travel Awards	2023
Delta Analytics Global Teaching Fellow	2021
GRIESHMA Research Scholar: Indian Institute of Technology–Chennai	2020
First Place Winner at Oracle’s Back to School Ethiopia Hackathon	2019

INVITED GUEST LECTURES AND TALKS

MetricMate: An Interactive Tool for Generating Evaluation Criteria for LLM-as-a-Judge	Fall 2025
Host: Zahra Ashktorab	
IBM Research	
OpenAI API Workshop	Spring 2025
CSE 40748/60748: Human-AI Collaborative Systems	
Host: Toby Li	
University of Notre Dame	
Prototyping with Figma Workshop	Spring 2025
CSE 40748/60748: Human-AI Collaborative Systems	
Host: Toby Li	
University of Notre Dame	
AI Assisted Qualitative Data Analysis	Spring 2024
EMR 6480: Qualitative Research Methods	
Host: Beixi Li	
Western Michigan University	

Design in the Margins

Fall 2022

CSE40424: Human-Computer Interaction

Host: Diego Gomez-Zara

University of Notre Dame

SERVICES

Computer Vision and Graphics Research Initiative 2026: Program Organizer	2026
ACM IUI 2026: Area Chair	2026
Reviewer: CHI 2023–2026, IUI 2024–2025, UIST 2024	2023–2026
ICLR 2025 Workshop on Bidirectional Human-AI Alignment: Program Committee	2025
University of Notre Dame Graduate Society of Women Engineers: Social Chair	2024
ACM IUI 2025: Social Co-Chair	2024
Student Volunteer: NeurIPS 2020, ACM FaccT 2022, ACM CHI 2023, ACM CHI 2024	2020–2024
INSPIRE - Addis Ababa University - Partnership Outreach Chair	2019–2020
Student Organizer for the First AI in Ethiopia Conference	2018

LANGUAGES & SKILLS

Programming Languages: Python, Java, Javascript, SQL, Kotlin**UX Skills:** Qualitative Research, Experiment Design, User Studies**Database:** MySQL, NoSQL, Oracle**Languages:** English—Fluent, Tigrigna—Native, Amharic—Native